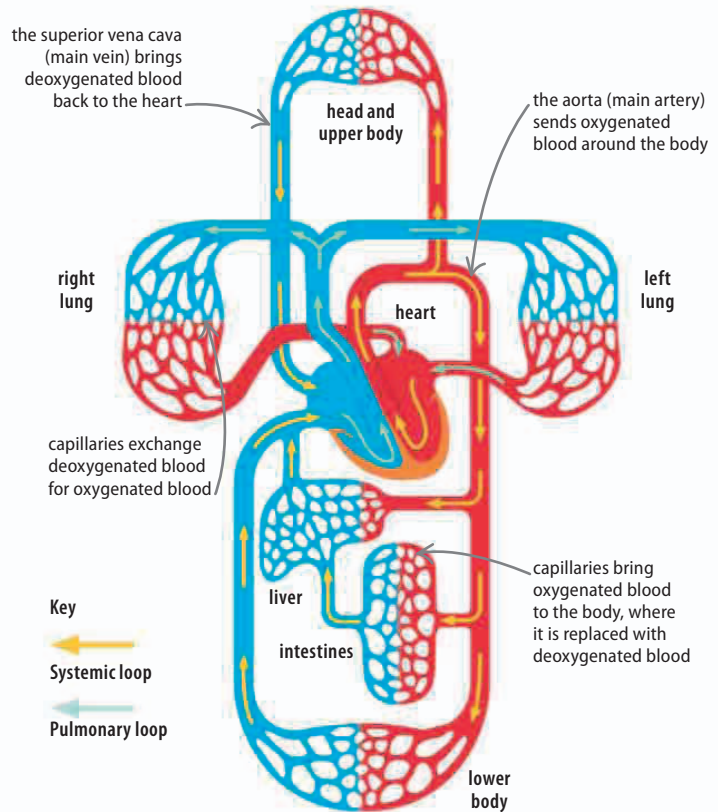


Circulatory system

The human circulatory system is a double loop of vessels. The pulmonary loop carries deoxygenated blood to the lungs, where it picks up oxygen and releases carbon dioxide. The reoxygenated blood then goes back to the heart, where it enters the second loop, the systemic loop, which takes it around the body.

▷ Vessel types

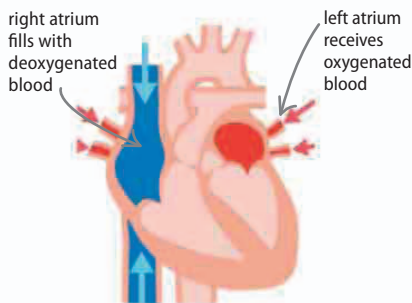
The arteries (in red) take oxygenated blood to the tissues. The system of veins (in blue) then brings back the used, deoxygenated blood—which is then returned to the lungs. Capillary vessels run between the arteries and veins, carrying blood through the tissues.



The **total length** of your circulatory system stretches an amazing 96,600 km (60,000 miles)—more than **twice** the distance around Earth.

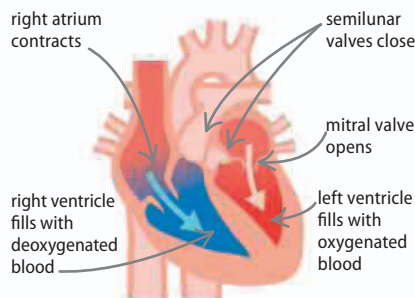
In a heartbeat

The human heart is a powerful pump made from a type of muscle (cardiac) that never needs to rest—so a heart can keep working throughout a person's life. The heart has two sides, each one divided into an upper chamber called the atrium, and a lower chamber (the ventricles). The right side receives deoxygenated blood from the body. Reoxygenated blood is pumped out again from the left side.



△ Heart relaxed

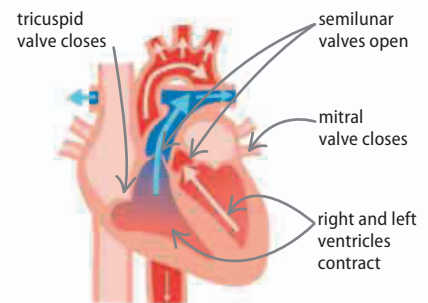
When the cardiac muscles are relaxed, deoxygenated blood flows into the right atrium from the vena cava, the main vein. Oxygenated blood flows into the left atrium.



△ Atria contract

The contraction of the heart starts at the top, squeezing the atria, so the blood moves down into the ventricles. One-way valves prevent the blood from moving back into the atria.

The heart beats around **three billion times** in the average person's life.



△ Ventricles contract

The lower part of the heart contracts, squeezing the ventricle. The right ventricle pumps the blood toward the lungs. The left ventricle pushes blood into the aorta (main artery).